Simon Gabriel

Mathematical Researcher at the University of California, Berkeley

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Focus

I am interested in mechanistic interpretability and its role in AI safety. In particular, I want to study neural networks as scale-dependent systems and ask how their internal representations and computational mechanisms emerge as model size and training change. My research develops probabilistic models of interacting systems in mathematical physics. I am beginning to explore how these methods can provide tractable models of neural networks and their behaviour.

Target Roles

I am available now for part-time research work. From January 2027, I am also available for a short research fellowship or a full-time position.

Relevant Experience

- **Mathematical research**
  I hold the Charles B. Morrey Jr. Math Fellowship at UC Berkeley. My record comprises nine papers and preprints, including three solo-authored works, in probability, stochastic analysis, and mathematical statistical physics.
- **AI-assisted mathematical research**
  I integrate AI agents into active mathematical research to explore proof strategies and possible approaches to open problems of interest to me. This work has generated new proof ideas and research directions.
- **Earlier technical work**
  Alongside my studies, I held part-time roles in data science, software development, and quantitative analysis, and completed a project-management internship at Siemens in Singapore. I worked on image recognition, data pipelines, ETF and index analysis, and portfolio algorithms.

Engagement with AI Safety

- **BlueDot Impact Technical AI Safety course**
  September–October 2026
- **Fields Academy Shared Graduate Course: Mathematics for AI Safety**
  September–December 2026

Background and Skills

Current strengths	Mathematical modelling; stochastic analysis; research writing; teaching.
Earlier technical experience	Python; C#; Java; MATLAB; SQL.